MATH-1014 T10

Tutorial7: Mixed Integration Strategies

WANG Yixiao

Department of Mathematics

March 23, 2026

Comprehensive Problem

Part III

(a) & (b)

([25 points]) Let $I_n = \int_0^{\frac{\pi}{3}} \tan^n \theta \sec^3 \theta \, d\theta$, where $n = 0, 1, 2, \ldots$ is an integer.

(a) By writing $\tan^n \theta \sec^3 \theta = \tan^{n-1} \theta \cdot (\tan \theta \sec^3 \theta)$ and using integration by parts, show that for $n \geq 2$,

$$I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}$$

[10 pts]

(b) Evaluate I_0. [4 pts]
(Hint: You may need the integral of $\sec \theta$*.)*

Solution to (a) - Part 1

Show: $I_n = \dfrac{8(\sqrt{3})^{n-1}}{n+2} - \dfrac{n-1}{n+2} I_{n-2}$

Let $u = \tan^{n-1}\theta \implies du = (n-1)\tan^{n-2}\theta \sec^2\theta \, d\theta$.
Let $dv = \tan\theta \sec^3\theta \, d\theta \implies v = \frac{1}{3}\sec^3\theta$.
Apply Integration by Parts ($\int u\,dv = uv - \int v\,du$):

$$I_n = \left[\frac{1}{3}\tan^{n-1}\theta \sec^3\theta\right]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \frac{1}{3}\sec^3\theta \cdot (n-1)\tan^{n-2}\theta \sec^2\theta \, d\theta$$

1. Evaluate the boundary term: At $\theta = \frac{\pi}{3}$, $\tan(\frac{\pi}{3}) = \sqrt{3}$ and $\sec(\frac{\pi}{3}) = 2$.
At $\theta = 0$, $\tan(0) = 0$ (since $n \geq 2$).

$$\text{Boundary} = \frac{1}{3}(\sqrt{3})^{n-1}(2)^3 - 0 = \frac{8}{3}(\sqrt{3})^{n-1}$$

Solution to (a) - Part 2

2. Simplify the integral term: Replace $\sec^2\theta$ with $1+\tan^2\theta$:

$$I_n = \frac{8}{3}(\sqrt{3})^{n-1} - \frac{n-1}{3}\int_0^{\frac{\pi}{3}} \tan^{n-2}\theta\sec^3\theta(1+\tan^2\theta)\,d\theta$$
$$I_n = \frac{8}{3}(\sqrt{3})^{n-1} - \frac{n-1}{3}\Big(I_{n-2}+I_n\Big)$$

Multiply the entire equation by 3 to clear denominators:

$$\begin{aligned} 3I_n &= 8(\sqrt{3})^{n-1} - (n-1)I_{n-2} - (n-1)I_n \\ (3+n-1)I_n &= 8(\sqrt{3})^{n-1} - (n-1)I_{n-2} \\ (n+2)I_n &= 8(\sqrt{3})^{n-1} - (n-1)I_{n-2} \end{aligned}$$

Divide by $(n+2)$ to obtain the final formula:

$$\boxed{I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2}I_{n-2}}$$

Solution to (b)

(b) Evaluate I_0:

$$I_0 = \int_0^{\frac{\pi}{3}} \sec^3\theta \, d\theta = \frac{1}{2}\Big[\sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|\Big]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{2}\left(2\sqrt{3} + \ln(2+\sqrt{3})\right) - 0 = \boxed{\sqrt{3} + \frac{1}{2}\ln(2+\sqrt{3})}$$

Problem 1: The LCM Rationalizing Substitution

Exercise 1

Evaluate the integral:

$$\int \frac{1}{\sqrt{x} + \sqrt[3]{x}}\, dx$$

Strategic Hint:
You have $x^{1/2}$ and $x^{1/3}$. To eliminate BOTH fractional powers, let $x = u^n$ where n is the Least Common Multiple of 2 and 3.
After substitution, check if you need polynomial long division!

Solution to Problem 1

1. Substitution:
Let $x = u^6$. Then $dx = 6u^5\,du$. Also, $\sqrt{x} = u^3$ and $\sqrt[3]{x} = u^2$.

$$\int \frac{1}{u^3+u^2} \cdot 6u^5\,du = \int \frac{6u^5}{u^2(u+1)}\,du = 6\int \frac{u^3}{u+1}\,du$$

2. Long Division:
Since degree 3 $\geq$ degree 1, we divide u^3 by $(u+1)$:

$$\frac{u^3}{u+1} = \frac{(u^3+1)-1}{u+1} = (u^2-u+1) - \frac{1}{u+1}$$

3. Integration & Back-Substitution:

$$6\int \left(u^2 - u + 1 - \frac{1}{u+1}\right) du = 6\left(\frac{u^3}{3} - \frac{u^2}{2} + u - \ln|u+1|\right) + C$$
$$= \boxed{2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\ln|\sqrt[6]{x}+1| + C}$$

Problem 2: Hidden Trig Substitution & Case III

Exercise 2

Evaluate the integral:

$$\int \frac{\cos x}{\sin^3 x + \sin x}\, dx$$

Solution to Problem 2

1. Substitution:
Let $u = \sin x$, then $du = \cos x\, dx$.

$$\int \frac{1}{u^3 + u}\, du = \int \frac{1}{u(u^2+1)}\, du$$

2. Partial Fractions:

$$\frac{1}{u(u^2+1)} = \frac{A}{u} + \frac{Bu + C}{u^2+1}$$

Comparing coefficients: $A = 1, C = 0, A + B = 0 \implies B = -1$.

$$\frac{1}{u(u^2+1)} = \frac{1}{u} - \frac{u}{u^2+1}$$

3. Integrate:

$$\int \left(\frac{1}{u} - \frac{u}{u^2+1}\right) du = \ln|u| - \frac{1}{2}\ln(u^2+1) + C$$

$$= \boxed{\ln|\sin x| - \frac{1}{2}\ln(\sin^2 x + 1) + C}$$

Problem 3: Factoring by Grouping & Completing the Square

Exercise 3

Evaluate the integral:

$$\int \frac{3x^2 - 2x + 5}{x^3 - x^2 + 2x - 2}\, dx$$

Solution to Problem 3

1. Factor & Partial Fractions:
Denominator: $x^3 - x^2 + 2x - 2 = (x-1)(x^2+2)$.

$$\frac{3x^2 - 2x + 5}{(x-1)(x^2+2)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+2}$$

$3x^2 - 2x + 5 = A(x^2+2) + (Bx+C)(x-1)$.
Let $x = 1 \implies 6 = 3A \implies \mathbf{A = 2}$.
x^2 terms: $3 = A + B \implies \mathbf{B = 1}$.
Constant terms: $5 = 2A - C \implies 5 = 4 - C \implies \mathbf{C = -1}$.

2. Integrate:

$$\int \left(\frac{2}{x-1} + \frac{x-1}{x^2+2} \right) dx = \int \frac{2}{x-1}\, dx + \int \frac{x}{x^2+2}\, dx - \int \frac{1}{x^2+2}\, dx$$
$$= \boxed{2\ln|x-1| + \frac{1}{2}\ln(x^2+2) - \frac{1}{\sqrt{2}} \arctan\left(\frac{x}{\sqrt{2}}\right) + C}$$

Problem 4: Algebraic Manipulation (The Conjugate Trick)

Exercise 4

Evaluate the integral:

$$\int \sqrt{\frac{1-x}{1+x}}\,dx$$

Solution to Problem 4

1. Algebraic Manipulation:
Multiply by $\sqrt{\frac{1-x}{1-x}}$:

$$\int \sqrt{\frac{1-x}{1+x}} \cdot \sqrt{\frac{1-x}{1-x}}\, dx = \int \frac{1-x}{\sqrt{1-x^2}}\, dx$$

2. Split into Two Standard Integrals:

$$\int \frac{1-x}{\sqrt{1-x^2}}\, dx = \int \frac{1}{\sqrt{1-x^2}}\, dx - \int \frac{x}{\sqrt{1-x^2}}\, dx$$

3. Integrate: The first integral is standard arcsine. For the second, let $u = 1 - x^2$, $du = -2x\, dx$.

$$\int \frac{1}{\sqrt{1-x^2}}\, dx - \left(-\frac{1}{2}\int \frac{1}{\sqrt{u}}\, du\right) = \arcsin x + \sqrt{u} + C$$
$$= \boxed{\arcsin x + \sqrt{1-x^2} + C}$$

Problem 5: Substitution into Integration by Parts

Exercise 5

Evaluate the integral:

$$\int e^{\sqrt{x}}\,dx$$

Solution to Problem 5

1. Substitution:
Let $u = \sqrt{x}$. Then $x = u^2$ and $dx = 2u\,du$.

$$\int e^{\sqrt{x}}\,dx = \int e^u \cdot 2u\,du = 2\int ue^u\,du$$

2. Integration by Parts:
Use $\int w\,dv = wv - \int v\,dw$.
Let $w = u \implies dw = du$.
Let $dv = e^u\,du \implies v = e^u$.

$$\begin{aligned} 2\int ue^u\,du &= 2\left(ue^u - \int e^u\,du\right) \\ &= 2(ue^u - e^u) + C \end{aligned}$$

3. Back-Substitute:

$$\boxed{2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C}$$

Problem 6: Completing the Square

Exercise 6

Evaluate the integral:

$$\int_1^3 \frac{4}{\sqrt{4x^2 - 8x + 13}}\, dx$$

(A) $\ln 3$ (B) $\ln 6$ (C) $\ln 7$ (D) $\ln 8$ (E) $\ln 9$

Solution to Problem 6

1. Complete the Square:

$4x^2 - 8x + 13 = 4(x^2 - 2x) + 13 = 4(x^2 - 2x + 1) - 4 + 13 = 4(x - 1)^2 + 9.$

$$\int_1^3 \frac{4}{\sqrt{4(x-1)^2+9}}\,dx = \int_1^3 \frac{4}{2\sqrt{(x-1)^2+\frac{9}{4}}}\,dx = 2\int_1^3 \frac{1}{\sqrt{(x-1)^2+\frac{9}{4}}}\,dx$$

2. Integrate:

Let $u = x - 1 \implies du = dx$. Limits: $u(1) = 0, u(3) = 2$.
Recall: $\int \frac{1}{\sqrt{u^2+a^2}}\,du = \ln\left|u + \sqrt{u^2+a^2}\right|$. Here $a^2 = \frac{9}{4}$.

$$I = 2\left[\ln\left|u + \sqrt{u^2+\frac{9}{4}}\right|\right]_0^2$$

3. Evaluate Bounds:

$$I = 2\left(\ln\left(2 + \sqrt{4+\frac{9}{4}}\right) - \ln\left(0 + \sqrt{0+\frac{9}{4}}\right)\right)$$
$$= 2\left(\ln\left(2+\frac{5}{2}\right) - \ln\left(\frac{3}{2}\right)\right) = 2\left(\ln\left(\frac{9}{2}\right) - \ln\left(\frac{3}{2}\right)\right)$$

Problem 7: Substitution into By-Parts

Exercise 7

Evaluate the integral:

$$\int_0^4 \ln(\sqrt{x}+1)\,dx$$

(A) $3\ln 3$ (B) $2\ln 3$ (C) $\ln 3$ (D) $\frac{3}{2}\ln 3$ (E) $\frac{3}{4}\ln 3$

Solution to Problem 7

1. Substitution:
Let $u = \sqrt{x}$. Then $x = u^2 \implies dx = 2u\,du$. Limits: $u(0) = 0, u(4) = 2$.

$$\int_0^4 \ln(\sqrt{x}+1)\,dx = \int_0^2 \ln(u+1)\cdot 2u\,du$$

2. Integration by Parts:
Use $\int w\,dv = wv - \int v\,dw$.
Let $w = \ln(u+1) \implies dw = \frac{1}{u+1}\,du$. Let $dv = 2u\,du \implies v = u^2$.

$$\begin{aligned}\int_0^2 2u\ln(u+1)\,du &= \left[u^2\ln(u+1)\right]_0^2 - \int_0^2 \frac{u^2}{u+1}\,du \\ &= 4\ln 3 - \int_0^2 \frac{u^2}{u+1}\,du\end{aligned}$$

3. Polynomial Division & Evaluate:
$\frac{u^2}{u+1} = \frac{u^2-1+1}{u+1} = u - 1 + \frac{1}{u+1}$.

$$\int_0^2 \left(u - 1 + \frac{1}{u+1}\right)\,du = \left[\frac{1}{2}u^2 - u + \ln|u+1|\right]_0^2$$